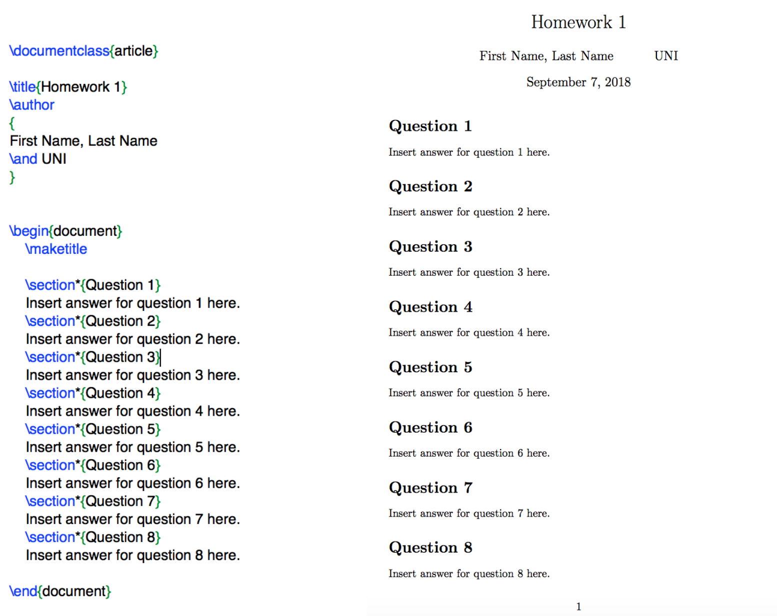
Homework 1: Discrete Mathematics

Introduction to LaTeX

COMS W3203

1 Introduction to LaTeX

LaTeX is a markup language that can be used to create well formatted PDF documents. It is widely used in science and academia. In this homework you will be asked to learn about LaTeX, use it to generate a very simple PDF document and submit your source code together with the resulting PDF. Here you can see an example (provided to you) of some LaTeX markup (`hw1_template.tex`) and the document it generates (`hw1_template.pdf`).



```
\documentclass{article}

\title{Homework 1}
\author
{
First Name, Last Name
\and UNI
}


\begin{document}
   \maketitle

   \section*{Question 1}
   Insert answer for question 1 here.
   \section*{Question 2}
   Insert answer for question 2 here.
   \section*{Question 3}
   Insert answer for question 3 here.
   \section*{Question 4}
   Insert answer for question 4 here.
   \section*{Question 5}
   Insert answer for question 5 here.
   \section*{Question 6}
   Insert answer for question 6 here.
   \section*{Question 7}
   Insert answer for question 7 here.
   \section*{Question 8}
   Insert answer for question 8 here.

\end{document}
```

Homework 1

First Name, Last Name UNI

September 7, 2018

Question 1

Insert answer for question 1 here.

Question 2

Insert answer for question 2 here.

Question 3

Insert answer for question 3 here.

Question 4

Insert answer for question 4 here.

Question 5

Insert answer for question 5 here.

Question 6

Insert answer for question 6 here.

Question 7

Insert answer for question 7 here.

Question 8

Insert answer for question 8 here.

1

2 Homework submission

In this homework, you will learn how to format mathematical expressions in LaTeX. For instance, to type the formula

$$f(x) = (x + y)^2$$

The associated LaTeX syntax is

```
\[f(x)=(x+y)^2\]
```

You can achieve a similar result using:

```
\centerline{$f(x)=(x+y)^2$}
```

Or using:

```
$$f(x)=(x+y)^2$$
```

3 Gradescope Submission

In this homework, you are expected **to submit two files:**

- A `.tex` file with the contents of your source code.
- A `.pdf` file with the results.

You'll see two assignments in Gradescope to submit both files. **It's important you submit both files.**

Make sure that you assign to each of the questions the corresponding page(s) from your PDF. If you're unsure what this means, here's a quick video tutorial to help: https://www.youtube.com/watch?v=u-pK4GzpId0

4 Frequently Asked Questions

1. Please respect all alignments in all questions. If an expression is centered, or justified left, please reproduce as is.
2. Please reproduce the formulas just like you see them.
3. You can use LaTeXpackages, like amsmath, or others.
4. Please use the template we gave you. Don't change the margins.
5. If you are using overleaf, you can download a PDF.
6. You can also download your project in overleaf. The zip file you get will contain a TEX file that you need to submit along with the PDF.
7. Yes, make sure to submit both files: your .tex and your .pdf.
8. Name your files hw1_uni.tex and hw1_uni.pdf where uni is your Columbia uni.
9. When you compile, **you can ignore any warnings but please fix the errors.**
10. Please be reminded that using LATEX or another typesetting program is highly encouraged for homework 2 onward. For this first homework, LaTeX is mandatory.
11. Now go to the next page and reproduce them using LaTeX! The resulting PDF should contain the following: a title, your name and UNI, the date on which it was generated, and answers to the questions below. Please use the given template `hw1_template.tex`.

    Have fun with LaTeX!

Question 1

A Pythagorean triple consists of three positive integers a, b, and c, such that

$$a^2 + b^2 = c^2$$

Write the condition $a^2 + b^2 = c^2$ in $\LaTeX$.

Question 2

Given a quadratic equation $ax^2 + bx + c = 0$ where x is an unknown variable, a, b, and c are constants. The solution to the quadratic equation is called quadratic formula and is given by:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Write the quadratic formula in $\LaTeX$.

Question 3

Use $\LaTeX$to write the following formulas:

$$1 + 2 + 3 + 4 + \cdots + 98 + 99 + 100 = \sum_{x=1}^{100} x$$

$$x_1 + x_2 + x_3 + \cdots + x_n = \sum_{i=1}^{n} x_i$$

$$x_1 \times x_2 \times x_3 \times \cdots \times x_n = \prod_{i=1}^{n} x_i$$

$$f(x) = \int_a^b x^2 dx$$

Question 4

This semester, we will learn about functions. Use $\LaTeX$to write the following function:

$$f : \mathbb{Z} \mapsto \mathbb{N}$$

$$f(x) = \begin{cases} 2x & \text{if } x \geq 0 \\ -2x - 1 & \text{if } x < 0 \end{cases}$$

Question 5

We will learn about set theory in this course. Use $\LaTeX$to write the following sets:

$$V = \{x \in \mathbb{Z} | x < 100\} \cap \{x \in \mathbb{Z} | \text{ x is prime}\}$$

$$V \subset W$$

Question 6

We will also learn about Boolean formulas and logic. Use LaTeXto write the following Boolean formula:

$$((\alpha \to \beta) \wedge (\beta \to \gamma)) \to (\alpha \to \gamma)$$

Question 7

Consider the following statements about integers:

1. For every x, there is a y, such that $x + y = 0$
2. There is a y, such that for every x, we have $x + y = 0$

Use LaTeXto write the translation of these two statements in symbols as follows. Please only reproduce the following two expressions, and make sure to reproduce the enumeration.

1. $\forall x\ \exists y\ x + y = 0$
2. $\exists y\ \forall x\ x + y = 0$

Question 8

Use LaTeXto write the following proof verbatim (as you see it below) including the square, mark of end of proof (did you see the square?). Hint: you may include a package to do that.

Proposition: If x is even, then x^2 is even.

Proof. x is an even number.
$\exists a \in \mathbb{Z}$ such that $x = 2a$
$x^2 = (2a)^2 = 4a^2 = 2(2a^2)$
Let $c = 2a^2$, $c \in \mathbb{Z}$
$x^2 = 2c$
Therefore, x^2 is even. $\square$

Question 9

Use LaTeXto write the following truth table:

x	y	$x \vee y$
TRUE	TRUE	TRUE
TRUE	FALSE	TRUE
FALSE	TRUE	TRUE
FALSE	FALSE	FALSE